Gabay — Shipping a Public RAG Product

One-page reflection · Graded Mini Project 14.3 · 2026-10-11 · gabay-fint.vercel.app · github.com/jacobvillar/gabay

Overview

Gabay is a public, domain-specific RAG assistant that answers questions about **Philippine statutory worker benefits** — 13th-month pay, leave, SSS, PhilHealth, and Pag-IBIG — strictly from official government handbooks, with the source passages shown under every answer. It is built with **Next.js + the Vercel AI SDK** (streaming chat, retrieval exposed as a tool call), **Groq** (open-weight GPT-OSS 120B) for generation, and **Upstash Vector** (built-in embedding model) for retrieval, deployed publicly on Vercel.

Corpus and key decisions

Decision	Choice	Why
Corpus	DOLE Handbook, SSS Citizens Charter, PhilHealth Konsulta circular, Pag-IBIG briefer	Public-domain PH government works — legal to serve publicly; a domain where base LLMs are unreliable, so retrieval + citations add real value
Generation	Groq — open-weight GPT-OSS 120B	Free tier, no OpenAI key, supports tool calling, deploys on Vercel
Embeddings	Upstash built-in model	Removes the embedding API key entirely — one fewer secret
Chunking	Per-page (PDF ligature repair) → sentence/paragraph-aware packing to ~1000 chars, 150 overlap	Per-page parsing preserves real **page numbers**; sentence-aware packing keeps rules intact
Metadata	`document_title`, `agency`, `page`, `source_url`	Readable citations that disambiguate across four agencies
Retrieval	Hybrid: dense + BM25 keyword, reciprocal rank fusion, optional per-agency boost; fallback chat model; two-search cap	Keyword catches exact terms dense misses; the agency boost counters the SSS skew; free-tier limits (10k writes, 200k tokens/day) degrade answers instead of breaking them
Interface	Onyx-style sidebar: New chat, collapsible Documents, Recents saved in the browser	No accounts or server storage — history stays on the visitor's device. Deliberately no admin upload: on a public app it would let anyone poison the index; new docs go through `npm run seed`

The trickiest decision

Making retrieval **trustworthy on high-stakes questions**. A wrong peso figure or rule is worse than no answer, so retrieval had to be a deliberate **tool call** (not always-prepended context), with a tool description tuned so the model reaches for it on benefit questions but skips small talk, and a system prompt kept strict: answer only from retrieved passages, and say plainly when the handbooks do not cover something. Balancing that faithfulness against still being genuinely helpful was the hardest part to get right.

What it does well

Grounded, cited answers on PH-specific rules. "How is 13th-month pay computed?" triggers the retrieval tool, answers from the DOLE handbook (1/12 of basic salary, with the correct exclusions), and renders the exact source pages — something a raw model gets subtly wrong with no citations.

How I tested it

An end-to-end eval of 10 questions against production (all four agencies, Taglish, and an off-topic probe), reading every answer. It caught five real defects: replies that ended empty because the model kept searching, PDF text broken 2,316 times (“bene fi t”), figures taken from model memory instead of the passages, an off-topic question answered, and raw HTML in tables. Each fix was re-run against the same questions, and the seed now verifies itself with a known-answer probe (13th-month pay must return DOLE pp. 50–54).

What it does badly

Retrieval balance. Because the SSS charter dominates the index, some non-SSS queries can surface less-relevant SSS passages; structure-aware chunking keeps each passage coherent, and the per-agency boost helps when the model uses it, but unscoped searches still lean SSS. These are quality, not correctness, issues — but they are real. The free tiers are the other weak spot: about 10-15 full answers a day before the smaller fallback model takes over, and it is noticeably less faithful.

What I'd do with another week

1. **Rebalance the corpus** — trim the SSS charter to its benefits sections (the per-agency boost is in place).
2. **Cross-encoder reranking** — rerank the fused dense + keyword results before they reach the model.
3. **Automate the eval** — turn the 10-question run into a scored harness (faithfulness, context relevance) that runs on every change.
4. **Inline citation mapping** — true [1]/[2] markers tied to the sources panel.